

Yi Cui

PhD Student in AI · Trustworthy Verification of Multimodal Reasoning & Agentic Systems

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EDUCATION

Ph.D. in Artificial Intelligence

2024 – Present

The Hong Kong University of Science and Technology (Guangzhou)

Supervisor: Prof. Hui Xiong — Founding Head of AI Thrust (HKUST-GZ); Fellow of ACM, IEEE, AAAS, AAAI, CAAI

M.S. in Computer Engineering (GPA 3.5/4.0)

2024

New York University

B.S. in Computer Science (GPA 3.4/4.0; Minor in Economics)

2021

Pennsylvania State University

RESEARCH

My research builds toward trustworthy, process-level verification of multimodal and agentic systems. Starting from benchmark construction that exposes how i.i.d. evaluation overstates real-deployment reliability (SafeBuild-Bench, KDD'26), I moved to the verification layer itself — diagnosing a hidden position-prior confounder that inflates multimodal process-reward scores (NeurIPS'26, under review). I am now extending accountability evaluation from perception to long-horizon agent behavior (RedDust), and mapping 100+ papers from visual chain-of-thought to verifiable reasoning agents into a process-level verification agenda (survey in preparation).

PUBLICATIONS

† Co-first author * Corresponding author

- [1] **SafeBuild-Bench: A Temporal-Robust Construction Safety Benchmark with Graph-Enhanced Data Mining.** Yi Cui†, Zilin Wang†, Yijie Xu, Qianyi Cai, Huizai Yao, Shuai Jiang, Bingzhao Zhong*, Hui Xiong*. **ACM SIGKDD 2026 (KDD)**, Datasets & Benchmarks Track — Accepted.
- [2] **Beyond Step AUC: Hidden Position Priors and Capability Trade-offs in Multimodal Process Reward Models.** Yi Cui, Huizai Yao, Zilin Wang, Zihan Wang, Hui Xiong*. **NeurIPS 2026** — Under Review.
- [3] **Beyond Boundaries: Leveraging Vision Foundation Models for Source-Free Object Detection.** Huizai Yao, Sicheng Zhao, Pengteng Li, Yi Cui, Shuo Lu, Weiyu Guo, Yunfan Lu, Yijie Xu, Hui Xiong*. **AAAI 2026** — Accepted.
- [4] **Gated Test-Time Scaling for GUI Grounding via Multi-Seed Zoom Search.** Huizai Yao, Yijie Xu, Tianfu Wang, Yili Wang, Yi Cui, Yunfan Lu, Fengjie Zhu, Xuemin Zhao, Hengyao Bao, Ying Sun*, Hui Xiong*. **ACM MM 2026** — Under Review.
- [5] **Contrastive Graph Matching for Domain Adaptive Object Detection.** Pengteng Li, Huizai Yao, Yi Cui, Yunfan Lu, Yijie Xu, Minghao Yang, Pinhao Song. **arXiv** — Preprint.

PROJECTS

Construction-Site Safety-Hazard VLM (Guangdong Heavy Industry, Industry Collaboration)

2025 – Present

Qwen3-VL-32B student (LoRA) · Qwen3-VL-235B-A22B teacher (distillation) · Contrastive labeling · GEMS · LLaMA-Factory · 8×A6000 · Agentic RAG · vLLM

- Pilot VLM system that flags construction-site safety hazards, cites the violated regulation, and drafts rectification advice — led the data pipeline, model adaptation, and evaluation end to end.
- Raised hazard-recognition accuracy by ~35% (relative) over baseline on the internal validation set across v2–v5 iterations — driven mainly by data-quality refinement (semantic extraction over expert annotations into a sharper hazard taxonomy) and system-prompt engineering (per-category definitions + a fixed output schema).
- Distilled a same-family Qwen3-VL-235B-A22B teacher into a Qwen3-VL-32B student via LoRA — the largest student our compute budget supports — using a contrastive teacher process to auto-generate structured “description →

hazard → regulation → fix” labels and sharply cut manual labeling cost.

- Curated an expert-reviewed 30K-instance SFT dataset from 100K+ raw inspection records, using GEMS (graph-enhanced data selection) to counter severe class long-tail and focus senior-engineer review on high-value data instead of the full corpus; expanded the hazard taxonomy from 62 to 90 classes along the way.
- Designing a decoupled Agentic RAG pipeline (in progress) to replace direct VLM generation of regulation citations, targeting traceability and maintainability.

RedDust — LLM-Agent Accountability & Trust Benchmark

Independent Project

TypeScript (deterministic headless engine) · React/Phaser · Anthropic Claude + DeepSeek agents · LLM-as-judge

- A 12-day survival-shelter benchmark that scores LLM agents not just on whether they survive, but on whether they stay accountable and do the right thing under pressure — an offline AI companion (AURA) advises but never decides, so accountability for every choice rests with the agent under test.
- Designed a 3-axis evaluation framework — outcome/resources, auditability (decision transparency, evidence, protecting weaker actors), and narrative/ethics (correctness independent of resource optimization).
- Built a deterministic, leak-controlled engine where one `runScenario()` powers both the React/Phaser visualization and headless benchmarking, with pluggable Claude/DeepSeek agents and offline LLM-as-judge grading of free-text “understanding probes” (a 2×2 matrix separating genuine understanding from lucky guessing and akrasia).
- Controlled ablations over one LLM + 4 scripted baselines (13 runs) decompose its long-horizon failure into two gaps: multi-day resource lookahead — closed by a projection scaffold (0%→100% win, matching a hand-coded expert agent) — and strategic branch selection, which explicit hints fail to close; the scoring axes are demonstrably orthogonal (greedy baseline: survival 100, principle 0; the LLM: principle/comprehension ≈1.0, win 0%).

SKILLS

Research Areas: Vision-Language Models (VLM/MLLM) · Multimodal Evaluation & Benchmarking · Process Reward Models (PRM) / Verifiable Reasoning · Data-Centric AI

Training & Methods: LoRA / PEFT · Knowledge Distillation · Contrastive Learning · Distributed Training (multi-GPU) · Agentic RAG

Programming: Python · C++ · Java **Frameworks/Tools:** PyTorch · MATLAB · React/Phaser · TypeScript · vLLM · LLaMA-Factory

TEACHING

Teaching Assistant, HKUST (Guangzhou)

AIAA 3053 — Reinforcement Learning: Principles and Methods

UFUG 2104 — Applied Statistics

LANGUAGES

English (Fluent — 6 years of study in the U.S.)

Chinese (Mandarin, Native)